

Optimizing Energy Savings and Operation of Active Distribution Networks Utilizing Hybrid Energy Resources and Soft Open Points: Case Study in Sohag, Egypt

M. B. SHAFIK^{1,2}, G. I. RASHED¹, AND HONGKUN CHEN¹

¹School of Electrical Engineering and Automation, Wuhan University, Wuhan 430072, China

²Electric Power System and Machines Department, Faculty of Engineering, Kafrelsheikh University, Kafrelsheikh 33516, Egypt

Corresponding authors: M. B. Shafik (mohamed.shafeeq@eng.kfs.edu.eg) and G. I. Rashed (ghamgeen@whu.edu.cn)

This work was supported in part by the National Key Technologies Research and Development Program of China under Grant 2018YFB0904303.

ABSTRACT This research develops an overall framework for optimal operation of Hybrid Renewable Energy Resources (HRERs) system that involves the combination of PV/WT/Diesel units. In the proposed framework, a co-simulator tool is proposed to find the optimal scheduling of HRERs parameters. It composes HOMER PRO and NEPLAN packages that are managed by C++ based online optimizer. Different scenarios of hybridization of energy resources are considered achieving technical, economic and environmental perspectives using the HOMER PRO package. The best allocation and sizing of Soft Open Points (SOPs) and the sharing of HRERs are obtained via multi-objective Artificial Electric Field Algorithm (AEFA). The simulation results propose a new hybrid grid-connected system to supply a realistic case study in Sohag-Egypt. That proposed HRERs system demonstrated a feasible economic solution at acceptable levels of gases emission and fuel consumption. Besides, the optimal operation of the HRERs system with SOPs potentially achieve significant technical perspectives that enhance the system losses, improve the voltage profile and increase the loading capacities of transmission lines. The technical perspective is handled using NEPLAN software. Furthermore, the simulation studies prove the high flexible planning and operation of distribution system at significant improvements technical, economic and environmental perspectives.

INDEX TERMS Active distribution networks, hybrid renewable energy resources, techno-economic analysis, HOMER PRO, NEPLAN, artificial electric field algorithm.

I. INTRODUCTION

The entire energy consumption is going to be increased by 30% in 2040 [1]. But about 75% of energy production comes from fossil fuel-based energy resources [2] that faces a lot of obstacles and result in a lot of problems [3, 4]. Renewables share in the electricity area will have rapid growth, as it is expected to provide almost 30% of power demand by 2023, while it was 24% in 2017 which is statistically stated by international energy agency (IEA) [5]. Also, the International Renewable Energy Agency (IRENA) forecasted that the global renewable energy share will reach 85% of the total energy supply by 2050 [2]. All of Renewables benefits and advantages doesn't reflect its low share in correspondence of

its low reliability level when it works as isolated source of energy due to nature of such resources [6]. Therefore, HRERs are developed as a lucrative solution for more environmental and technical issues associated with electrical energy market.

About renewable energy potential in Egypt; it possesses outstanding wind profile with average wind speeds between 7 and 10 m/s and 2,000 to 3,000 kWh/m²/year of direct solar radiation [3]. However; only 0.38% of total energy consumption comes from renewables but the Egyptian government plan targets to increase the total installed capacity of renewables to 11% by 2020 and up to 20% by 2030 [7].

A. SCHEDULING OF HRERs: LITERATURE REVIEW

HRERs system is known as a combination between renewable energy resources with some conventional sources (diesel generator) along with storage units, in order to satisfy the

The associate editor coordinating the review of this manuscript and approving it for publication was Shiwei Xia¹.

demand of specific area in reliable, economic and environmental manners [8].

So far, many categories of research are available in the literature in the field of hybrid energy systems that covered design, planning, and energy management subjects. Most of these studies are used a simulator named Hybrid Optimization Model for Electric Renewables (HOMER) [8-10]. The HOMER PRO (PRO denotes to the software edition) is used to optimize energy savings through a techno-economic system electrification plan. A comprehensive survey about HRERs is presented in [3]; this survey mentioned using HOMER software for designing the energy supply plan in different countries (Somali, Iran, Colombia, Malaysia, Us). HOMER succeeded to achieve all of these studies objectives. For instance, in Yamen study to supply electricity for a rural area, while they proposed PV/WT/Diesel/Battery isolated system. This attained around 45% and 70% reduction in the total Cost of Energy (COE) and CO₂ emissions compared with deiseal system. Similar study but this time in Hurgada - Egypt where, a substantial decrease in COE from 0.444 \$/kWh (using diesel) to 0.139 \$/kWh, as well as decreasing the total emitted CO₂ from 461,667 kg/yr. to 78,500 kg/yr. All research in literature especially in Egypt give a push to get a good hybridization plan for system in Sohag to get all new system benefits.

B. ACTIVE DISTRIBUTION NETWORK

The concept of Active Distribution Networks (ADNs) is initiated from passive networks that are designed to distribute the received bulk power from the main grid to loads. Moreover, ADNs are provided with new technologies for active/reactive power support, advanced Distribution Management System (DMS) for supervision, control, operation, planning of distributed HRERs and Microgrids [11]. The research work in [12] provides a comprehensive comparison of the most recently proposed losses management approaches and strategies especially from the perspectives of losses minimization. The authors show that the optimal allocation of distributed HRERs scheme is the best solution for minimum system losses and a more reliable and stable operation. Within the same trend, the authors of [13] used quasi-static time-series for optimal allocation of Distributed Generation (DG) and static transformers for economic and technical considerations of unbalanced distribution networks and they succeeded to achieve higher systems efficiency compared to normal operation. NEPLAN software is used to simulate active distribution networks and perform a lot of studies such as load flow, short circuit and reliability analysis.

In the direction of ADNs modelling, the authors of [14] proposed a nonlinear model of ADNs which is validated with standard test system and proved capabilities of the proposed model in response to system disturbances; However, that model needs modifications to deal with problems of interface with the main grid. While the authors in [11] present a summary and assessment of the uncertainty modeling techniques, concluding that there is no single best uncertainty modeling

technique yet, but selection of the best technique depends on the problem and the type of uncertain input variables. Some researches focused on the role of power electronic devices like SOPs in ADNs especially reactive power management capabilities and increasing the hosting capacity of DG [15], [16]. While in [17] a study of optimal allocation of DG synchronized with capacitor banks and SOPs in ADNs is presented but without considering a multi-objective study, economic selection of DG units, increasing the penetration level of DG units. Moreover, the authors of [17] showed the GA results for a low number of iterations which is not enough to avoid falling into a local optimum point.

C. OPTIMIZATION ALGORITHMS DIVERSITY

Plenty of research work in literature provides a comprehensive survey about various optimization techniques especially those are used in solving power system problems is [18: 21]. AEFA is an intelligent population-based optimization algorithm that enthused its principal of operation from Colom's law of the electrostatic force and Newton's law of motion [20]. AEFA uses the mechanism of charged particles where best agent fitness has the largest charge and attracts all other agents with a specific force and acceleration velocity towards an optimal solution. AEFA also benefits from personal and global best positions to learn other particles to fasten reaching optimal solution. None of the available researches in the literature used AEFA to investigate electric power system problems, therefore this research proposed AEFA algorithm as an efficient and new optimization technique that can handle the multi-objectives and nonlinearities in the power system.

D. PAPER CONTRIBUTION AND ORGANIZATION

The contributions of this paper are briefed in the following points:

- Proposing a co-simulator for techno-economic planning and optimal operation of ADNs.
- A novel application of the new intelligent optimization technique (AEFA) for power system problem solution.
- Using SOP capabilities to increase DG penetration level for better voltage stability and provide a procedure for finding the optimal DNs reconfiguration accomplished with DG allocation and SOP switching action for better energy utilization efficiency.
- Applying the proposed methodology on a real system application and benefited from its location in planning HRERs from economic considerations. On the other hand, technical considerations are considered through optimal operation study for the new system.

The reminder of the paper is organized as follows: Section 2, prevails the objectives and work methodology which consists of Co-simulator, AEFA optimizer working procedures, techno-economic planning, optimal operation model, SOP model, HRERs model of Homer and radial DN for NEPALN. A detailed discussion of obtained simulation results is

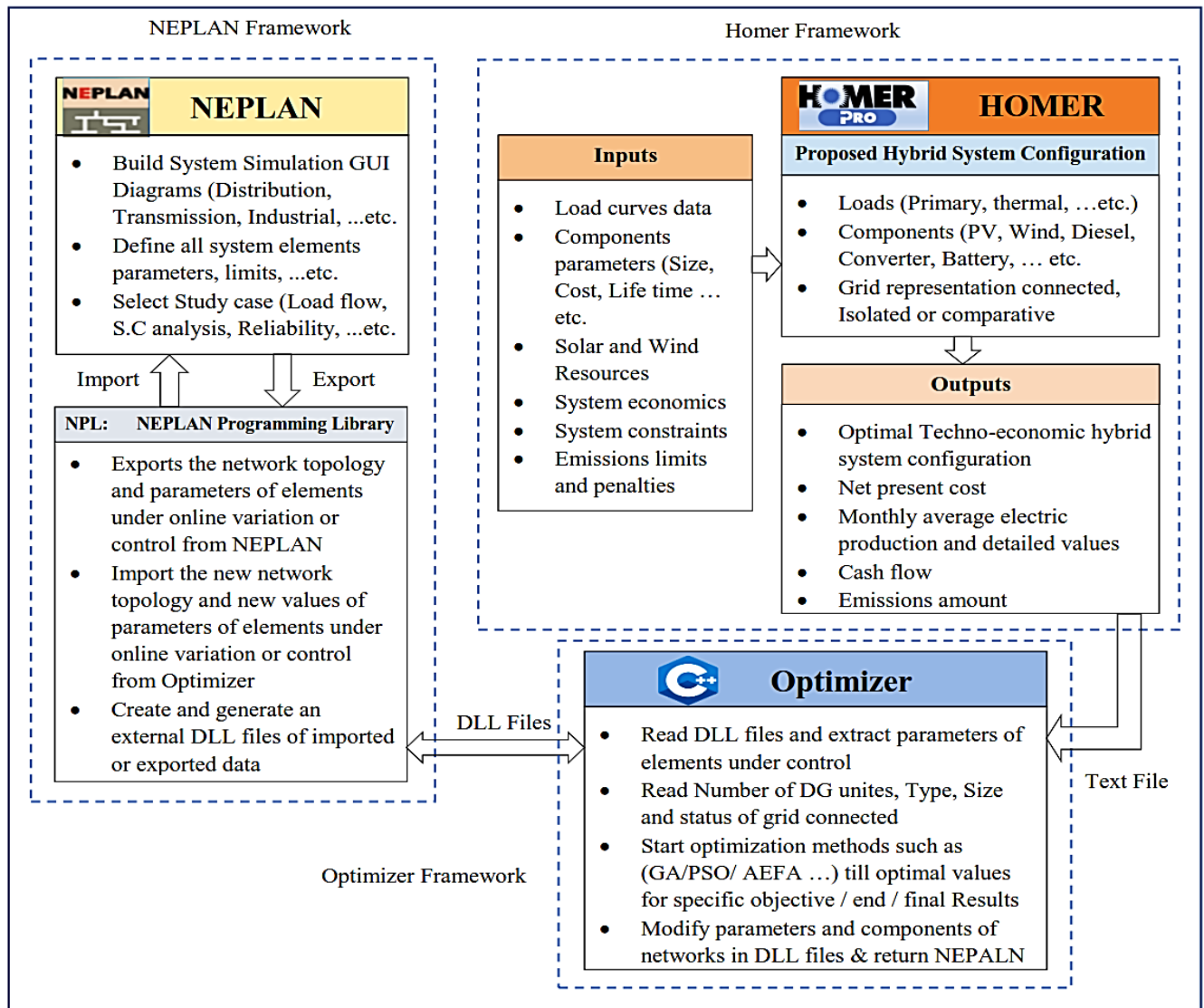


FIGURE 1. Holistic frame work of co-simulator for power system studies.

presented in section 4. Finally, conclusions are drawn in section 5.

II. RESEARCH OBJECTIVES AND METHODOLOGY

A. RESEARCH OBJECTIVES

Two main objectives are considered in the proposed approach. *The first* is providing a sustainable and clean supply to the total load demand of AND of Sohag using a hybrid PV/WT/Diesel/Converter grid-connected renewable energy system. To reach this target, a feasibility study of hybrid system implementation is performed. The study based on techno-economic analysis to define the best configuration plan of the hybrid grid-connected system. *The second* objective is increasing the HRERs energy utilization efficiency. To attain that goal a multi-objective optimal power flow study to select the best locations and size for the selected renewables for better system performance. Moreover, the distribution

network is equipped with SOPs as a new technology that helps in enhancing system performance indices in normal and emergency conditions.

B. ANTICIPATED CO-SIMULATOR

To achieve the paper objectives a co-simulator composed of HOMER and NEPLAN simulators with controlling Optimizer is proposed. Fig. 1 illustrates how this co-simulator works. *HOMER PRO* is the most trusted, robust and widely used tool to simulate, optimize and do sensitivity investigation for HRERs system isolated or connected to grid [9]. Homer like any optimization tool defines its objectives (economic factors, practicality minimizing NPC and COE) and put some criteria and assumptions then search in the possible configurations to find the best plan. Sohag network elements like nodes and branches is simulated in detail without any simplifications using *NEPLAN* through an efficient

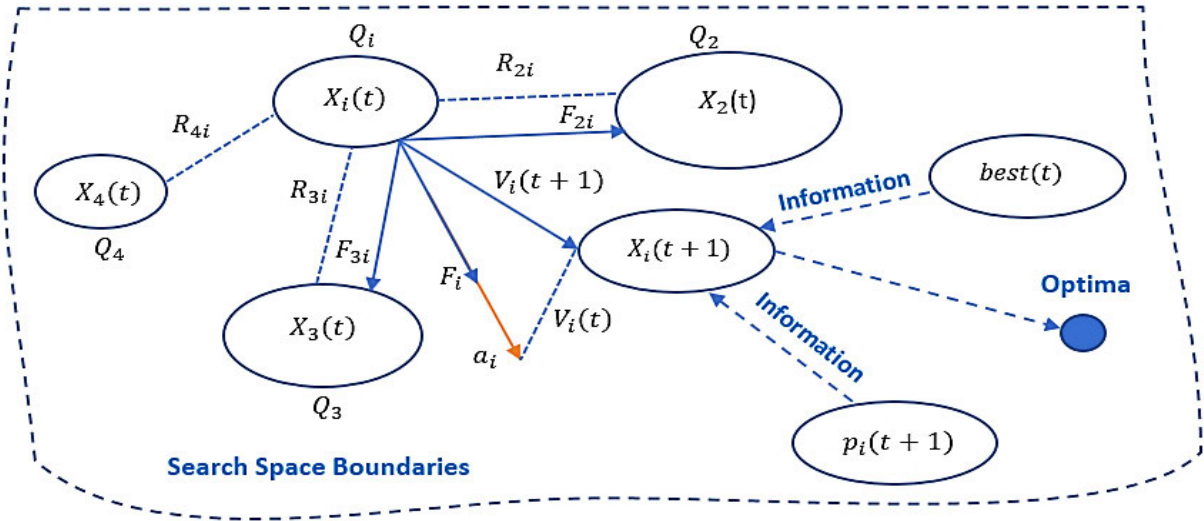


FIGURE 2. Concept of AEFA searching mechanism.

graphical user interface. Then the *online optimizer* receives the HOMER's plan (keeping it fixed) export system parameters and topology then starts to find the optimal allocation for HRERs units, SOP devices. Then the online optimizer performs multiple trials to reach optimal system operation. The online optimizer is always ready for any load changes or in system configuration or even in HOMER set plan and starts immediately to find best operating conditions. Thanks to NEPLAN Programming Library (NPL) which is used to export all system-controlled variables to Optimizer. Then uses a C++ artificial electric field algorithm (AEFA) [20] for Optimal power flow framework with a set of iterations till reaching best performance.

C. ARTIFICIAL ELECTRIC FIELD ALGORITHM (AEFA)

As mention in the introduction, AEFA is a population-based heuristic algorithm. AEFA identify its population as a charged particle which attract or repel each other causing a directed motion. The charges (controlled variables) are a function of the fitness (objective function). The greatest charged particle ("best" individual) attracts all other particles of lower charge and move towards the objective. The AEFA algorithm can be explained in the following subsections:

1) POPULATION

Each particle can be expressed as $x_i = \{x_i^1, x_i^2, \dots, x_i^d \forall i = 1, 2, \dots, N\}$; where x_i^d is the i^{th} particle position of d -dimension (fitness function variables) search space while, the fitness function $fit(x_i^d(t))$ of each particle i of each variable d at any iteration t .

2) CHARGE ($Q_i^D(T) = Q_j^D(T)$) AND LEARNING STRATEGY

Fitness function is used to determine the charge of each particle ' $fit(x_i^d(t))$ ' depending on best, worst and current value of each particle (Eq. 1). Maximization and

minimization process depend on monotonically increasing or decreasing respectively, which assuring the largest charge and hence greater force for the best objective value as in Eq.2.

$$q_i^d(t) = \exp\left(\frac{fit((x_i^d(t))) - worst(t)}{best(t) - worst(t)}\right), \quad (1)$$

where $best = \text{Min}(fit(x_i^d(t)))$

$worst = \text{Max}(fit(x_i^d(t)))$

$$Q_i^d(t) = Q_j^d(t) = \frac{q_i^d(t)}{\sum_{i=1}^N q_i^d(t)} \quad \forall i, j = 1, 2, \dots, N \quad (2)$$

3) FORCE $F_{ij}^d(t)$ AND SHARING INFORMATION OF PERSONAL BEST

Eq.3 describes the force acted by charge j on charge i at iteration t . The Euclidian distance between two particles i and j given by expression 4 as a norm function. The Coulomb's constant is taken as a function of iteration (t) and the max. iteration, it can be calculated by Eq. 5 where k_o and α are constants, while, ϵ is a small positive constant and $rand$ is a random number. Then Eq.6 expresses resultant electric force acting on the i^{th} particle as in Fig.2 that illustrates how AEFA particles search for an optimal solution into the search space limits.

$$F_{ij}^d(t) = k(t) \frac{Q_i^d(t) * Q_j^d(t) * (p_i^d(t) - x_i^d(t))}{R_{ij}^d(t) + \epsilon} \quad \forall i, j = 1, 2, \dots, N \quad (3)$$

$$R_{ij}^d(t) = \|x_i^d(t) - x_j^d(t)\|_2 \quad (4)$$

$$k(t) = k_o * \exp(-\alpha \frac{t}{t_{max}}) \quad (5)$$

$$F_i^d(t) = \sum_{j=1, j \neq i}^N rand * F_{ij}^d(t), \quad rand \in [0, 1] \quad (6)$$

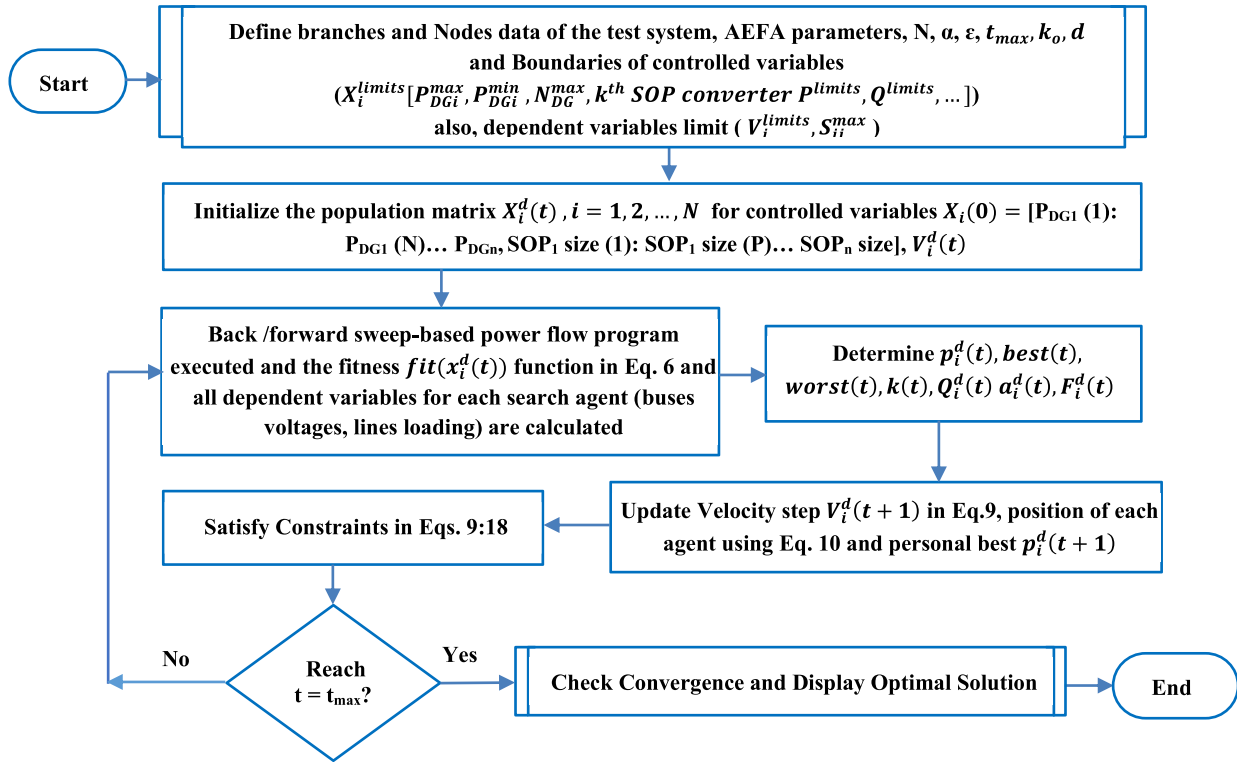


FIGURE 3. AEFA algorithm procedure for power flow framework.

4) VELOCITY STEP FUNCTION $V_i^d(t)$

The velocity step is initially estimated within limits and then updated from Eq. 9 where the electric field of the i^{th} particle at any time t and in d^{th} dimension is given by Eq. 7. When $M_i^d(t)$ is the unit mass of the i^{th} particle then each particle acceleration can be calculated from Eq.8.

$$E_i^d(t) = \frac{F_{ij}^d(t)}{Q_i^d(t)} \quad (7)$$

$$a_i^d(t) = \frac{E_i^d(t) * Q_i^d(t)}{M_i^d(t)} \quad (8)$$

$$V_i^d(t+1) = V_i^d(t) * rand + a_i^d(t) \quad (9)$$

5) UPDATE AGENTS' POSITION

After sharing information of personal and global best solution; the step velocity is used to update the current position of each agent to continue search for the optimal value as in Eq. 10. The position of the best fitness value obtained by any particle i at any iteration t is also updated by the Eq. 11. The flow process of AEFA optimizer and the combination between optimization and power flow is depicted in Fig. 3.

$$X_i^d(t+1) = X_i^d(t) + V_i^d(t+1) \quad (10)$$

$$p_i^d(t+1) = \begin{cases} p_i^d(t) & \text{if } fit(p_i^d(t)) < fit(x_i^d(t+1)) \\ x_i^d(t+1) & \text{else} \end{cases} \quad (11)$$

D. TECHNO-ECONOMIC MODELLING

After HRERs system data entry and simulation, HOMER explores all possible hybrid system arrangements then offer set of feasible and optimal solutions. The first main objective of this research work is to minimize the total system NPC and COE subject to technical and environmental restrictions. From Eq.12, the total system NPC is a function of total system annualized cost (C_{t_an}) and the capital discounting factor (DF). Eq.13 represents the total system components annual cost as the summation of capital, replacement and operation, and maintenance (O&M) costs of system components with specific life time. DF is calculated from Eq.14, as a function of the discount rate (interest rate) 'D' on capital investments of the installed equipment, and the project lifetime. Finally, the COE is given as in equation (15), where E_{t_an} is the total annual useful energy generated from the HRERs system to source the electrical load.

$$NPC = \frac{C_{t_an}}{DF} \quad (12)$$

$$C_{t_an} = \sum_{k=1}^n (C_{capital,k} + C_{repl,k} + C_{O\&M,k}) \quad (13)$$

$$DF = \frac{D(1+D)^p}{[(1+D)^p - 1]} \quad (14)$$

$$COE = \frac{C_{t_an}}{E_{t_an}} \quad (15)$$

E. MULTI-OBJECTIVE POWER FLOW PROBLEM FORMULATION

The second main objective is to improve the performance of the resulted HRERs system of Sohag. The online optimizer minimizes the system active power losses and reduce total voltage deviation with adequate DG penetration level 'PL'. The online optimizer considers set of control variables that include locations of DG units of HRERs system and its sharing level. Also, SOP site and sizing (Reactive power value absorbed or injected by SOP / SOP active power flow value and direction).

To minimize losses and voltage deviation a multi-objective function is presented in Eq. (16), with the help of priorities weighting factors $W_1:W_3$. The first priority is to minimize system active power losses then reduce the total deviation of nodes voltage beyond nominal voltage for better voltage stability margin. Maximizing the sharing level 'PL' of HRERs system units helps in reducing system losses and voltage deviation so, the transpose of 'PL' is added to the objective formula last section.

$$Obj = Min \left(W_1 * P_{Losses} + W_2 * \sum_{i \in Nf} |V_n - V_i| + W_3 * \frac{1}{PL} \right) \quad (16)$$

$$PL = \left(\sum_{i \in NDG} P_{DG_i} \right) / P_{Gt}, \text{ Where } P_{Gt} = \sum_{i \in NDG} P_{DG_i} + P_{slack} \quad (17)$$

$$P_{Losses} = \sum_{i,j \in Nf} Real \left(I_{ij}^2 * Z_{lij} \right) \forall Zl \in Nl. \quad (18)$$

PL is expressed in Eq. (17) as the percentage of total DG generated power compared to the total power consumption by the system from DG units added to P_{slack} from the main grid. Network losses is the sum of all branches real power losses in Eq.18 while ' Z_{lij} ' is branch impedance and ' I_{ij} ' is current flow though the same branch. The aforementioned objective function in Eq. (16) is exposed to a set of operational constraints can be categorized and explained as:

1) TOTAL ACTIVE/REACTIVE POWER BALANCE

As stated in Eq. 19 The total power generation must satisfy system load demand and losses. Reactive power balance is restricted by Eq. 20 while system reactive power losses that can be calculated in terms of Eq.21; where Q_i^K and Q_j^K are kth SOP reactive power injected or absorbed at both sides of SOP of real power losses P_{Losses}^k .

$$P_{Gt} = \sum_{i \in Nf} P_{Di} + P_{losses} + P_{Losses}^k \quad (19)$$

$$Q_{Gt} = \sum_{i \in Nf} Q_{Di} + Q_{losses} + Q_i^K + Q_j^K \quad (20)$$

$$Q_{Gt} = \sum_{i \in NDG} Q_{DG_i} + Q_{slack} \quad (20)$$

$$Q_{Losses} = \sum_{i,j \in Nf} Imag \left(\mu_{ij} * I_{ij}^2 * Z_{lij} \right) \forall Zl \in Nl \quad (21)$$

2) NODAL CURRENT BALANCING CONSTRAIN

The first side of Eq. 22 represents the sum of all currents entering or leaving node i from or to connected nodes Nc added to current injected into bus i if it is provided DG units I_{DG_i} . That side must balance the current I_{load_i} which is drawn by the load at the same node.

$$\sum_{j \in Nc} *I_{ij} + I_{DG_i} = I_{load_i} \quad \forall i \neq j \quad (22)$$

3) ACTIVE/REACTIVE GENERATION LIMIT OF DG UNITS

P_{DG_i} and Q_{DG_i} should be selected according to the available ranges, so the objective function is also constrained with Eqs. 23-25 where the number of DG units is also limited to the available committed units N_{DG}^{max} :

$$P_{DG_i} \leq P_{DG_i}^{max} \quad \forall i \in NDG \quad (23)$$

$$Q_{DG_i} \leq Q_{DG_i}^{max} \quad \forall i \in NDG \quad (24)$$

$$NDG \leq N_{DG}^{max} \quad (25)$$

4) SECURITY CONSTRAINTS

Nodes voltage levels must be kept within limits also lines MVA loadings that can't exceed lines thermal loading S_{ij}^{max} as expressed in Eqs. (26) and (27) respectively. The power flow direction through the kth SOP installed between buses i, j can be controlled according to Eqs. 28 and 29 while SOP power flow can be controlled using operator ε_k that can be selected as 0, +1 or -1.

$$V_i^{min} \leq V_i \leq V_i^{max} \cdot i \in Nf \quad (26)$$

$$S_{ij} \leq S_{ij}^{max} \quad \forall i \neq j \& i \cdot j \in Nf \quad (27)$$

5) SOP CONSTRAINTS

The power flow direction through the kth SOP installed between buses i, j can be controlled according to Eqs. 28 and 29 while SOP power flow can be controlled using operator ε_k that can be selected as 0, +1 or -1. Eq. 30 shows that Power entering or leaving SOP nearly equal while SOP active losses is neglected. Also, reactive power entering and leaving SOP can be controlled with different values and different directions to represent the capability of reactive power absorption or compensation. But also, these values of reactive power must be limited to permissible converters limits $Q_{i,j}^{Kmin}, Q_{i,j}^{Kmax}$ of the kth SOP as in 31,32:

$$\varepsilon_k * P_{ij}^k \leq P_{ij}^{max} \quad \forall k \in N_{SOP} \& \left| P_{ij}^k \right| = \left| P_i^k \right| = \left| P_j^k \right| \quad (28)$$

$$\varepsilon_k = \begin{cases} 0 & \text{SOP}_{ij}^k \text{ disconnected/not exist} \\ +1 & \text{flow from bus i to j} \\ -1 & \text{flow from bus j to i} \end{cases} \quad (29)$$

$$P_i^k + P_j^k + P_{Losses}^k = 0; \quad k \in N_{SOP} \quad (30)$$

$$Q_{i,min}^K \leq Q_i^K \leq Q_{i,max}^K; \quad K \in N_{SOP} \quad (31)$$

$$Q_{j,min}^K \leq Q_j^K \leq Q_{j,max}^K; \quad K \in N_{SOP} \quad (32)$$

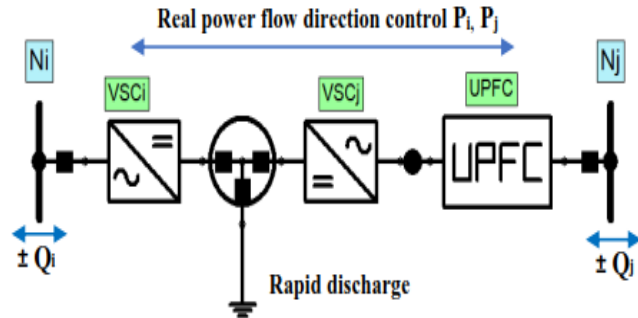


FIGURE 4. SOP modeling using NEPLAN.

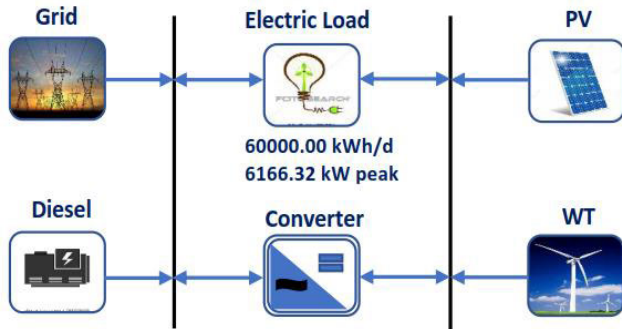


FIGURE 5. Schematic diagram of the proposed HRERs system of Sohag case study.

F. SOFT OPEN POINTS MODELING

SOP is defined as a multi-function power electronic device installed into DNs to replace Tie lines or any branch to be used as an active/reactive power flow control tool. SOP is modeled in NEPLAN as a static model as in Fig.4 while there are no dynamic considerations in this work focus.

SOP setup is Back-to-back Voltage Source Converter (VSC) series with unified power flow controller. For smooth control of voltage and active/reactive power flow in steady state converter rating is 10 MVA [25:28] with VSC/PWM PQ control mode. Besides, UPFC uses active/reactive power flow lines flow regulation with voltage regulation $\pm 5\%$ at both ends.

G. COMPONENTS OF THE PROPOSED HYBRID RENEWABLE ENERGY SYSTEM

The overall scheme of the hybrid renewable energy system is presented in Fig. 5. The technical and economic parameters of each component of Photovoltaic array, wind Turbine, D. generator and converter that are inserted in HOMER are mentioned comprehensively in [3]. Sohag system is always connected to grid so it is given to HOMER to be inserted in all possible plans. Grid tariff is 0.144 \$/kWh with a sell return price of 0.08 \$/kWh; also, emissions are 632.00 gm/kWh carbon dioxide [3].

H. SOHAG-EGYPT CASE STUDY

HOMER provides the best HRERs plan; then the online optimizer selects the best location for these units. HRERs best locations is presented in Fig. 6. Sohag network

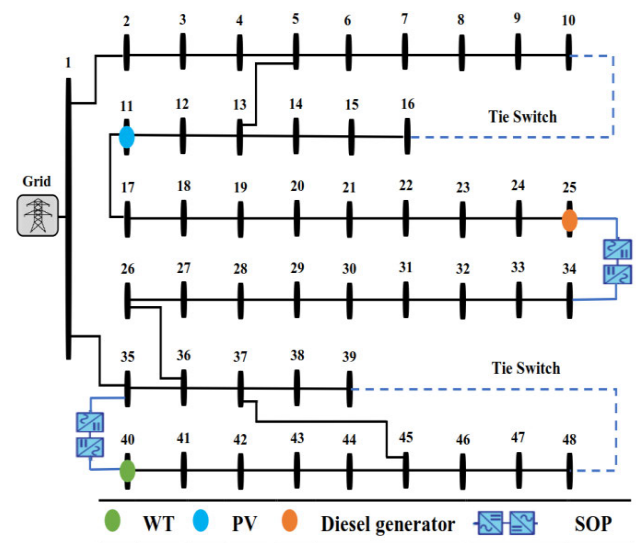


FIGURE 6. Sohag-Egypt active radial distribution network.

TABLE 1. AEFA parameters setting.

k_o	500	t_{max}	200	W_1	0.85
α	30	N	300	W_2	0.09
ε	2.22 E-16	d	12	W_3	0.06

consists of 48 load points linked with 47 branches and 4 tie lines. Moreover, optimizer also finds the best location for SOP devices that replaces tie lines between nodes 25, 34 and 35, 40.

Sohag system is balanced radial DN which is fed from one substation 22/11 kV (10 MVA transformer at Slack bus 1 and all loads at the low voltage side of transformers 11kV/400V). System peak load is 6154 kW and 3271 kVar which is distributed over all system nodes except the main bus connected to the grid. Site specifications and resources data are mentioned in [3]. The system voltage deviation limits are $\pm 5\%$. The cable used for DN branches is the same for economic considerations; which is (Appendix)12 kV – single core, XPLE insulated, PVC sheathed, but the cable from the grid to bus 1 is 22 kV- three core, XPLE/PVC steel armored conforming to IEC 60502-2 [22]. All the system detailed nodes, branches, cables and transformers mentioned in Appendix.

III. SIMULATION RESULTS AND DISCUSSION

A. TECHNO-ECONOMIC STUDY SCENARIOS AND OPTIMAL POWER FLOW STUDY CASES

For HOMER to cover all possible hybridization scenarios, eight different setups are simulated. Each scenario is evaluated based on techno-economic analysis methodology previously in sub-section c of section 3;. Where the optimal one is obtained with minimum NPC and COE with a reasonable emission level:

- Scenario 1: Grid-PV-WT-Diesel G.-Converter
- Scenario 2: Grid-PV-WT-Converter

TABLE 2. Techno-Economic optimized results of hybrid renewable energy system.

Case	System Configuration	Components Size				Cost				RF (%)	CO ₂ Emissions (kg/yr.)	Fuel (L/yr.)
		PV (kW)	WT (Qty.)	G (kW)	C (kW)	NPC (M\$)	COE (\$/kWh)	O&M (M\$/yr.)	Capital (M\$)			
1	Grid-PV-WT-G-C	2000	35	700	1500	34.3	0.117	1.9	9.67	39.4	8,909,810	1,220,409
2	Grid-PV-WT-C	2000	40	-	1500	34.8	0.177	1.92	9.97	42.2	8,410,011	0
3	Grid-PV-G-C	2000	-	800	1500	35.2	0.124	2.51	2.77	16.3	11,887,659	1,719,788
4	Grid-PV-C	2000	-	-	1500	36.1	0.128	2.64	1.97	16.3	11,586,400	0
5	Grid-WT-G	-	45	1000	-	38.3	0.13	2.19	10	30.4	10,296,618	1,576,236
6	Grid-WT	-	50	-	-	38.9	0.13	2.24	10	38.9	9,730,151	0
7	Grid-G	-	-	1000	-	39.9	0.141	3.01	1	0	14,177,239	1,922,647
8	Grid	-	-	-	-	40.8	0.144	3.15	0	0	13,840,800	0

- Scenario 3: Grid-PV- Diesel G. - Converter
- Scenario 4: Grid-PV- Converter
- Scenario 5: Grid-WT- Diesel G
- Scenario 6: Grid-WT
- Scenario 7: Grid- Diesel G
- Scenario 8: Grid

Concerning optimizer and NEPLAN for optimal system operation; four cases are employed to accentuate the contribution of this paper:

- Case 1: Normal Operation with Basic reconfiguration, while no DG or SOP introduced into system.
- Case 2: Still no DG or SOP introduced into system but only some system reconfiguration in normal operation.
- Case 3: Only DG unites is inserted without any reconfiguration or SOP.
- Case 4: the best topology, optimum DG penetration level besides also optimum size, control, and placement of SOP.

Some of study assumption is listed in [3] like project life time which is set to 25 years. The considered HOMER energy dispatch approach is based on the load following concept, which tends to be suitable with systems high penetration of renewables [16]. Using only two SOPs at most because of its high cost and it is not planned in that work focus. The current carrying of cable in NEPLAN is set to 250 A for more reserve in lines for future extensions. AEFA optimizer parameters setting for the multi-objective problem introduced in table 1.

B. OPTIMAL CONFIGURATION PLAN OF THE HYBRID GRID CONNECTED SYSTEM OF SOHAG

Table 2 gives the full simulation results of different arrangements scenarios with the help of the HOMER PRO software package. In all scenarios, the electricity production satisfy required load within the permissible rating of various system components. Table 2 shows the optimal size of the system component at each configuration that gives the cost data including NPC (\$) and COE (\$/kWh) besides, the value of the total system emission (kg/yr.).

Furthermore, the table introduces the value of renewable fraction (RF). RF can be defined by Eq. 33 as the percentage the energy delivered to the load from renewable power sources where the total electrical and thermal energy

produced from non-renewable sources E_{g_nonren} and H_{g_nonren} and the total electrical and thermal energy consumption E_{load} and H_{load} .

$$\%RF = \left(1 - \frac{E_{g_nonren} + H_{g_nonren}}{E_{load} + H_{load}}\right) \times 100 \quad (33)$$

Some highlights on the HOMER techno-economic results in table 2.:

- Scenario 1 gives the optimal cost-effective configuration plan of Sohag area case under study. Where it results in the minimum NPC (34.3 M\$) and COE (0.117 \$/kWh) with a moderate RF of 39.4%. The resulted HRERs system involves 2000 kW PV, a 25 WT with 100 kW rating each, 700 kW diesel generator in with a converter rating of 1500 kW.
- The optimal HRERs system reasonably reduces NPC and COE by about 16% and 18.75% compared to Grid only scenario with gas emission of 8,909,810 kg/yr. However, Optimal configuration reduces the total emissions by 35.65%, it is worth to appoint to that scenario 1 is the best scenario from the environmental perspectives with the maximum RF.
- scenario 8 with Grid generators only (zero RF) is the worst economic and environmental case scenario with NPC of 40.8 M\$, a defined COE tariff of 0.144 \$/kWh besides, the total gas emission goes up to its top value of 13,840,800 kg/yr.
- After the abovementioned highlights, it is settled that the HRERs system of Grid/PV/WT/G/C is the best setup plan for Sohag area as it gives a reasonable result regarding technical, economic and environmental aspects.

The monthly average electric share of each renewable source compared to the grid for optimal hybridization plan is graphically illustrated in Fig. 7. Table 3 presents energy generation of the optimal HRERs plan, Renewable resources (PV, WT) are the main sources to supply load by 9,197,139 kWh/yr. (40.1%) compared to the main grid of 39.4% and diesel (20.6%). As expected from the cheaper cost of PV besides the high value of average solar radiation in Sohag, PV has a high value of electricity generation but is limited with space for PV array which interprets the diesel share by in total energy production.

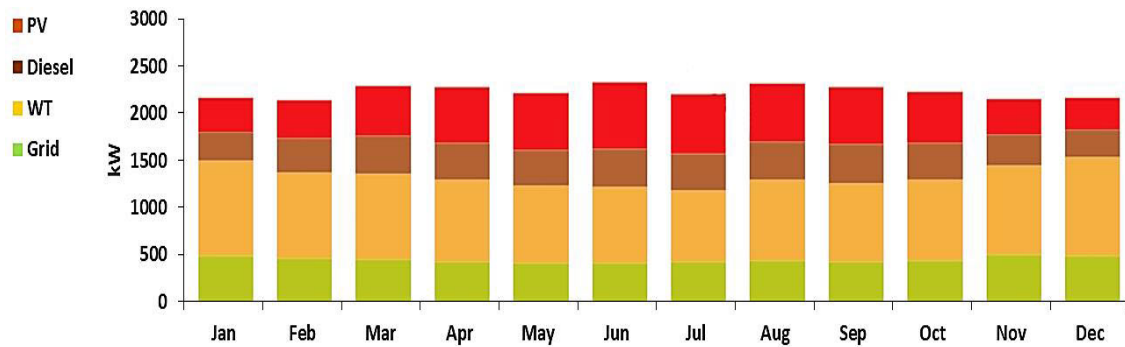


FIGURE 7. Monthly average electric production for optimal configuration plan (Scenario 1).

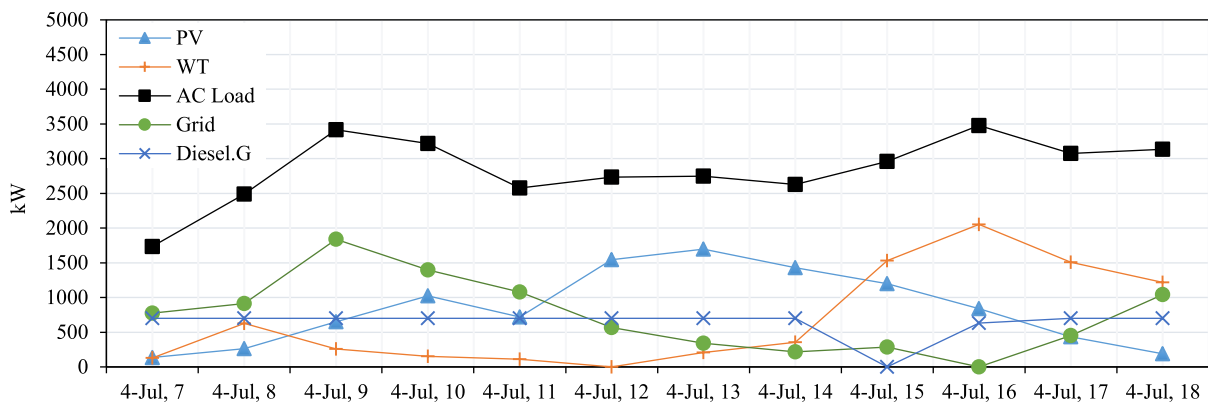


FIGURE 8. Hourly generated power for optimal configuration plan, (7:00 AM-18:00 AM, 4 July).

TABLE 3. Summary of the electrical energy production and consumption results for the optimal case scenario.

Source	(kWh/yr.)	%	Load	(kWh/yr.)	%
PV	3,804,888	16.6	Capacity Shortage	0	0
G	4,726,594	20.6	Unmet load	0	0
WT	5,392,251	23.5	Electric Load	21,9 E5	96.4
Grid	9,034,212	39.4	Grid Sales	820,172	3.61
Total	22,957,945	100	Excess Electrical	50,030	0.218

TABLE 4. Summary of the electrical energy production and consumption results for the optimal case scenario.

Component	Cost (M\$)					
	Capital	Replace	O&M	Fuel	Salvage	Total
PV	1.2	0	0.000259	0	0	1.2
G	0.7	0	2.417	5.206	-0.043	8.28
Grid	0	0	15.969	0	0	15.97
Converter	0.773	0	0	0	0	0.773
Wind	7	2.232	0.0679	0	-1.258	8.042
Overall System	9.673	2.232	18.455	5.206	-1.301	34.26

Regarding energy consumption, it is observed from table 3 that the optimal configuration plan succeeds to meet 24-hr load demand, see Fig.8 with zero capacity shortage, as well as a backup excess energy of 50,030 kWh/yr.

Regarding overall economic assessment, Table 4 includes capital, replacement, O&M, fuel, and salvage costs for system components and the overall system of the optimal configuration plan. The expensive capital cost of WT, results in the

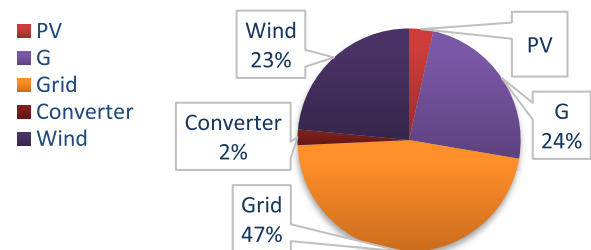


FIGURE 9. Distribution of the total cost of the overall system between system components (Scenario 1).

higher total cost between system components with 8.042 M\$ which approximately equals 23% of the total system cost.

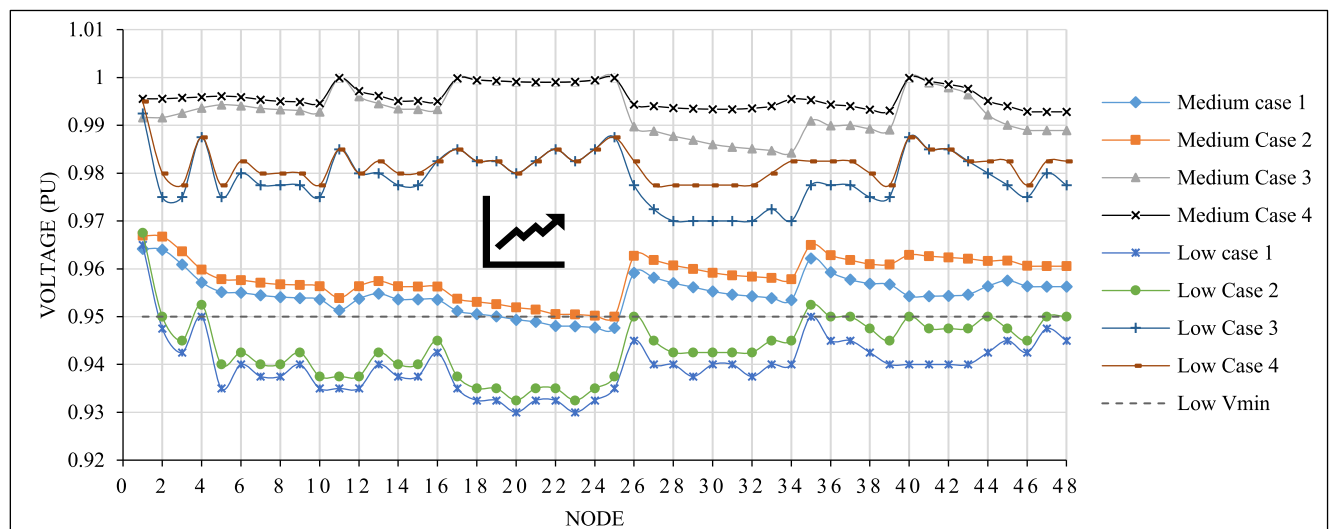
Likewise, diesel generator also shares by a substantial amount with 8.28 M\$, but in that case due to O&M and fuel cost. On the other hand, PV comes with approximately free O&M cost, so the PV array is considered as the cheapest component in the HRERs system. Fig. 9. Summarizes table 4 as it shows in a comparative graph the total system costs distribution between differentsystem components.

C. OPTIMAL OPERATION OF SOHAG ADN

After planning stage new Sohag ADN in Fig. 6 is provided with DG units and SOPs but without any sizing or control. So multi-objective optimization is performed using AEFA optimizer. Maximum load demand case of 6154 kW and 3271 kVar is selected to be fed with the new hybrid system

TABLE 5. Summary of the optimal power flow results for the four cases under study.

Case	DG			SOP control			Power losses			Voltage (PU)				% lines loading Decline	% PI
	Size (kVA)	Type	at bus	P (kW)	Q (kVar)	Between buses	P (KW)	Q (kVar)	Decline %	V _{min}	@ bus	Σ Div. to V _n	Σ Div. to V _{min}		
1	...	...	...	...	...	...	140.2	658.34	...	0.948	25	2.885	0.485	...	0
2	...	...	...	100	50 C	25-34	129.8	329.88	7.37	0.95	25	1.996	0.404	6.79	0
				700	600 C	35-40									
3	700	PV	11	...	...	...	83.23	183.23	40.61	0.984	34	2.065	0.334	18.55	33.91
	900	G	25												
	960	WT	40												
4	700	PV	11	800	500 C	25-34	60.11	166.36	57.11	0.993	48	2.205	0.195	48.68	36.05
	900	G	25												
	960	WT	40												

**FIGURE 10.** Low and medium voltage profile improvement of cases 1: 4.

and see the effect on system losses, lines load flow, and voltage profile.

Table 5 shows the results of the optimal power flow of sohad ADN. This table compares other cases with base case 1 of normal operation where losses are 140.2 kW and 658.34 kVar with a minimum voltage of 0.948 PU at medium bus 25 with large voltage deviation. with zero contribution of DG units; Case 2 shows two SOPs are put in place of tie switches between buses 25-34 and 35-40. SOPs reduced system losses from 140.2 kW, 658.34 kVar to 129.8 kW, 329.88 kVar with active losses decline of 7.37% while, SOPs have a small effect in system voltage profile.

Case 3 denotes to the presence of DG units only without SOPs so AEFA finds optimal locations for the DG units as in Fig.6 also maximize sharing value with the grid to feed the system load. In that case, suitable integration of DG units with a penetration level of 33.91% has a great effect on system active losses that have reduced with 40.6% compared to normal operation also, reactive power losses reduced from 658.34 kVar to 183.23 kVar. Also, integrating DG changes all the system power flow and results a good voltage profile where; minimum voltage of 0.948 PU at normal operation became 1 PU while in case 3 minimum voltage shifted to

bus 34 with 0.984 PU. This case 3 reduced total voltage deviation compared to the nominal value from 2.885 PU to 2.065 PU so it increased the voltage stability margin also.

In case 4 while using both DG and SOPs active power losses decreased by 57.11%. the optimizer used capabilities of SOPs to increase DG penetration level to 36.05 instead of 33.91 with case 3 without SOPs. It is noted also that SOPs reverse the direction of power flow in the tie line between nodes 35-40 while compensates reactive power to the system. Voltage profile improving is very important especially in lower voltage side directly connected to loads and that is clearly shown in Fig. 10.

Fig. 10 illustrates that both medium voltage and consequently low voltage profile is enhanced within limits; Especially in case 4 that proved capabilities of SOPs with DG in providing good voltage profile at all nodes. Moreover, they kept the low voltage side nodes within limits after the low voltage transformers action which proved that case 4 is the best for most efficient system performance. Normally when the proposed algorithm succeeded in minimizing losses; DN line loading currents is also decreased. Fig. 11 explains that the presence of DG and SOP reduces most of the lines loading

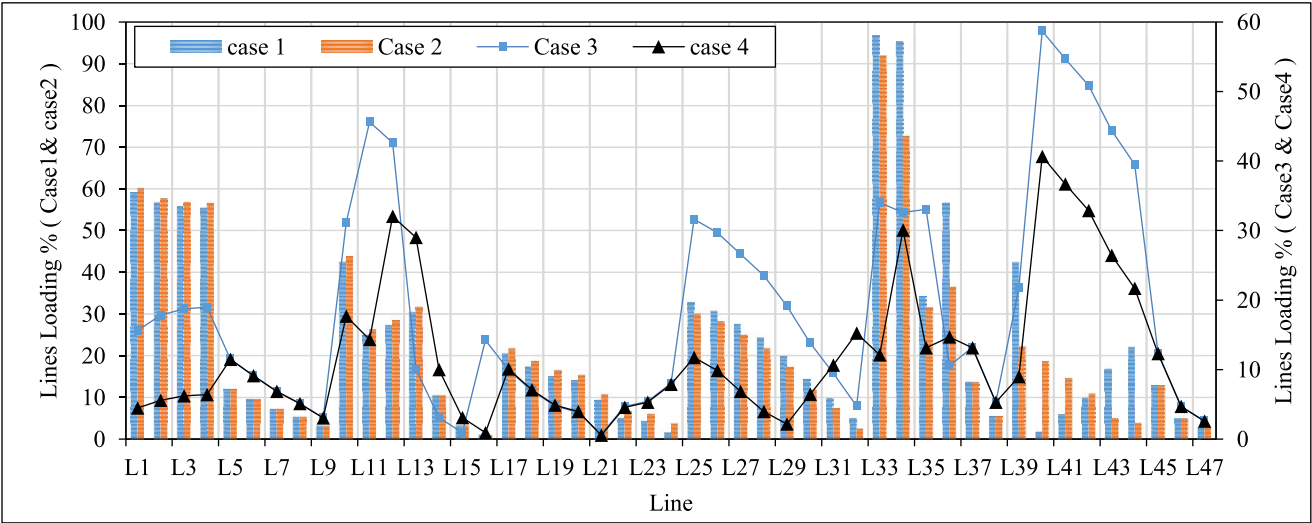


FIGURE 11. System branches loading comparison for cases 1: 4.

TABLE 6. Summary of the optimal power flow results for the four cases under study.

Function	Metric	Algorithm				
		PSO	ABC	WVO	ACO	AEFA
f_2	best	4.16E+04	3.38E+04	2.00E+02	3.79E+04	2.05E+02
	worst	1.07E+07	6.82E+04	2.89E+02	1.44E+07	1.25E+04
	Std	8.95E-08	7.28E-11	2.40E+01	1.05E+03	2.70E+03
f_4	best	4.38E+03	4.41E+03	4.39E+02	5.80E+03	4.07E+02
	worst	9.24E+02	8.26E+03	5.30E+02	9.79E+03	4.07E+02
	Std	0.00E+00	1.55E+02	1.93E+01	1.73E+02	1.37E-03
f_7	best	7.96E+02	7.02E+02	7.08E+02	7.01E+02	7.01E+02
	worst	8.37E+02	7.98E+02	7.15E+02	8.36E+02	7.01E+02
	Std	1.33E-11	3.20E+00	7.83E+00	1.03E+01	1.32E-04
f_{15}	best	4.23E+03	2.63E+03	1.60E+03	2.80E+03	1.60E+03
	worst	6.38E+03	3.18E+03	1.60E+03	5.93E+03	1.60E+03
	Std	1.41E-10	7.18E+00	2.23E-13	6.97E+01	2.10E-02

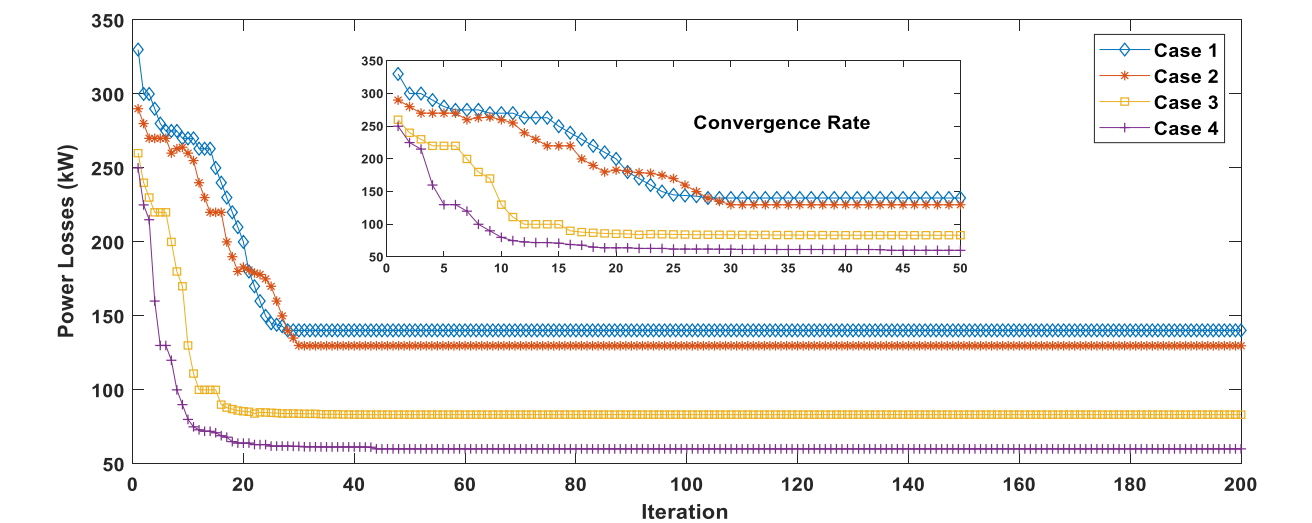


FIGURE 12. AEFA convergence for different cases 1:4

FIGURE 12. AEFA convergence for different cases 1:4.

especially in case 4 that reduced the total lines loading by 48.68% compared to 18.55% in case 3 which reflects also the role of SOPs in decreasing level of lines loading and increase system safety.

TABLE 7. Sohag system detailed data.

Feeders data					
Bus 1: Slack bus, no load, 11 kV- connected to grid					
Feeder	Node	ID	Loading		Voltage (kV)
			kW	kVAR	
Feeder 1	2	S1-01	97	47	11
	3	S1-02	40	19	11
	4	S1-03	10	5	11
	5	S1-04	40	19	11
	6	S1-05	100	49	11
	7	S1-06	94	45	11
	8	S1-07	74	36	11
	9	S1-08	87	42	11
	10	S1-09	127	62	11
Feeder 2	11	S2-01	92	45	11
	12	S2-02	128	62	11
	13	S2-03	64	31	11
	14	S2-04	291	141	11
	15	S2-05	92	45	11
	16	S2-06	43	5	11
Feeder 3	17	S3-01	179	87	11
	18	S3-02	126	61	11
	19	S3-03	90	50	11
	20	S3-04	40	20	11
	21	S3-05	190	90	11
	22	S3-06	170	80	11
	23	S3-07	30	20	11
	24	S3-08	110	50	11
Feeder 4	25	S3-09	60	30	11
	26	S4-01	60	30	11
	27	S4-02	80	40	11
	28	S4-03	130	60	11
	29	S4-04	130	60	11
	30	S4-05	180	90	11
	31	S4-06	220	110	11
	32	S4-07	180	90	11
Feeder 5	33	S4-08	200	90	11
	34	S4-09	200	90	11
	35	S5-01	60	30	11
	36	S5-02	190	90	11
	37	S5-03	30	10	11
	38	S5-04	330	160	11
	39	S5-05	220	110	11
Feeder 6	40	S6-01	70	30	11
	41	S6-02	170	80	11
	42	S6-03	160	80	11
	43	S6-04	260	170	11
	44	S6-05	190	140	11
	45	S6-06	290	190	11
	46	S6-07	270	230	11
	47	S6-08	70	70	11
	48	S6-09	90	80	11

Branches data						
Feeder	From	No	To	Length (m)	Voltage (kV)	Type
	Grid		Bus1	519	22	Type 3
Feeder 1	Bus 1	1	S1-01	420	11	Type 1
	S1-01	2	S1-02	320	11	Type 1
	S1-02	3	S1-03	365	11	Type 1
	S1-03	4	S1-04	245	11	Type 1
	S1-04	5	S1-05	415	11	Type 1
	S1-05	6	S1-06	350	11	Type 1
	S1-06	7	S1-07	920	11	Type 1
	S1-07	8	S1-08	415	11	Type 1
	S1-08	9	S1-09	510	11	Type 1
Feeder 2	S1-05	10	S2-03	50	11	Type 1
	S2-01	11	S2-02	115	11	Type 1
	S2-02	12	S2-03	395	11	Type 1
	S2-03	13	S2-04	365	11	Type 1
	S2-04	14	S2-05	305	11	Type 1
Feeder 3	S2-05	15	S2-06	230	11	Type 1
	S2-01	16	S3-01	425	11	Type 1
	S3-01	17	S3-02	400	11	Type 1
	S3-02	18	S3-03	230	11	Type 1
	S3-03	19	S3-04	440	11	Type 1
	S3-04	20	S3-05	1235	11	Type 1
	S3-05	21	S3-06	760	11	Type 1
	S3-06	22	S3-07	235	11	Type 1
	S3-07	23	S3-08	700	11	Type 1
Feeder 4	S3-08	24	S3-09	920	11	Type 1
	S4-01	25	S4-02	485	11	Type 1
	S4-02	26	S4-03	210	11	Type 1
	S4-03	27	S4-04	245	11	Type 1
	S4-04	28	S4-05	375	11	Type 1
	S4-05	29	S4-06	800	11	Type 1
	S4-06	30	S4-07	325	11	Type 1
	S4-07	31	S4-08	920	11	Type 1
	S4-08	32	S4-09	340	11	Type 1
Feeder 5	S4-01	33	S5-02	530	11	Type 1
	Bus 1	34	S5-01	420	11	Type 1
	S5-01	35	S5-02	245	11	Type 1
	S5-02	36	S5-03	375	11	Type 1
	S5-03	37	S5-04	800	11	Type 1
Feeder 6	S5-04	38	S5-05	395	11	Type 1
	S5-03	39	S6-06	600	11	Type 1
	S6-01	40	S6-02	210	11	Type 1
	S6-02	41	S6-03	245	11	Type 1
	S6-03	42	S6-04	375	11	Type 1
	S6-04	43	S6-05	230	11	Type 1
	S6-05	44	S6-06	440	11	Type 1
	S6-06	45	S6-07	1235	11	Type 1
	S6-07	46	S6-08	760	11	Type 1
Tie Lines	S6-08	47	S6-09	920	11	Type 1
	S1-09	48	S2-06	...	11	TS
	S3-09	49	S4-09	...	11	SOP
	S1-05	50	S1-06	...	11	SOP
	S5-05	51	S6-09	...	11	TS

D. AEFA OPTIMIZER ASSESSMENT

Statistical validation for AEFA optimizer is performed using IEEE CEC15 learning based benchmark test suite [20], the test is done on 15 different in nature functions. table 6 shows the results of four functions of each category; it's found

that AEFA produces outstanding performance compared to other techniques. It is worth to say that AEFA gives the best results in 7 functions out of the 15 and near results to best in other functions. The efficiency of AEFA optimizer concerning to the generations to reach an optimal solution

TABLE 7. (Continued.) Sohag system detailed data.

Cable Parameters		Cable category		
		Type 1	Type 2	Type 3
		1/C, CU/XLPE/PVC, 12 kV (25 kV) Cables	3/C, CU/XLPE/PVC, 12 kV (25 kV) Cables	3/C, CU/XLPE/PVC, 22 kV (35 kV) Cables
		300 kCMIL (150 mm ²)	300 kCMIL (150 mm ²)	300 kCMIL (150 mm ²)
Conductor DC resistance @ 20 °C (Ω/km)		0.115819	0.1158193	0.1158193
Conductor AC resistance @ 90 °C (Ω/km)		0.148301	0.1483012	0.1483012
Reactance inductive X (Ω/km)		0.127959	0.121397	0.1295995
Positive/negative sequence impedance Z (Ω/km)		0.193579	0.193579	0.19686
Zero sequence resistance R ₀ @ 90 °C (Ω/ km)		0.974457	0.974457	0.9081808
Zero sequence reactance X ₀ (Ω/ km)		0.547927	0.547927	0.4780417
Current Capacity (A)	Under ground	430	420	430
	In air	475	445	485
	In duct	395	375	410
Transformers Data				
ID	Node	Rating (MW)	%Z	X/R
TR0	Bus 1	10	4	13
CSS-1-XE108	2	0.2	4	3.09
CSS-1-XE109	3	0.08	4	2.47
CSS-1-XE113	4	0.05	4	2.47
CSS-1-XE109	5	0.08	4	2.47
CSS-1-XE117	6	0.25	4	3.09
CSS-1-XE108	7	0.2	4	3.09
CSS-1-XE116	8	0.16	4	3.09
CSS-1-XE108	9	0.2	4	3.09
CSS-1-XE117	10	0.25	4	3.09
CSS-1-XE108	11	0.2	4	3.09
CSS-1-XE117	12	0.25	4	3.09
CSS-1-XE116	13	0.16	4	3.09
CSS-3-XE118	14	0.63	4	3.96
CSS-1-XE108	15	0.2	4	3.09
CSS-1-XE109	16	0.08	4	2.47
CSS-1-XE111	17	0.4	4	3.09
CSS-1-XE117	18	0.25	4	3.09
CSS-1-XE108	19	0.2	4	3.09
CSS-1-XE109	20	0.08	4	2.47
CSS-1-XE111	21	0.4	4	3.09
CSS-1-XE111	22	0.4	4	3.09
CSS-1-XE109	23	0.08	4	2.47
CSS-1-XE117	24	0.25	4	3.09
CSS-1-XE116	25	0.16	4	3.09
CSS-1-XE116	26	0.16	4	3.09
CSS-1-XE116	27	0.16	4	3.09
CSS-1-XE117	28	0.25	4	3.09
CSS-1-XE117	29	0.25	4	3.09
CSS-1-XE111	30	0.4	4	3.09
ET3-116	31	0.5	4	3.09
CSS-1-XE111	32	0.4	4	3.09
ET3-116	33	0.5	4	3.09
ET3-116	34	0.5	4	3.09
CSS-1-XE116	35	0.16	4	3.09
ET3-116	36	0.5	4	3.09
CSS-1-XE109	37	0.08	4	2.47
CSS-3-XE117	38	0.8	5	5.79
ET3-116	39	0.5	4	3.09
CSS-1-XE116	40	0.16	4	3.09
CSS-1-XE111	41	0.4	4	3.09
CSS-1-XE111	42	0.4	4	3.09
CSS-3-XE117	43	0.8	5	5.79
CSS-4-XE117	44	0.63	4	3.96
CSS-1-XE114	45	1	5	5.79
CSS-1-XE114	46	1	5	5.79
CSS-1-XE111	47	0.4	4	3.09
CSS-1-XE111	48	0.4	4	3.09

as a new tool to solve Electrical engineering problems is assessed via convergence curves in Fig.12. AEFA reached optimal solution with a convergence rate of 43 iterations with a large number of controlled variables in case 4 which is rapid and accurate. To avoid that the tool falls into a local optimum; trails of solution continued for 200 iterations to prove the capability of the proposed algorithm has a good balance for 4 cases under study.

IV. CONCLUSION

In this research work the feasibility of Sohag-Egypt system electrification with HRERs (PV/WT/Diesel/Converter) grid-connected system has been investigated with the help of HOMER software. Besides, this hybridization plan is tested from the optimal operation standpoint using NEPLAN software; while the Online Optimizer works as a controller for the whole the process (from planning to optimal operation). Proposed hybrid system (2000 kW PV, 35×100 kW WT together with 700 kW diesel generator as well as a and 1500 kW system converter) succeeded in obtaining the minimum NPC, COE with the lowest gas emissions in addition to a reasonable value of RF compared to grid side. Good plan is not a guarantee for good operation so, optimizer aimed at finding the best size and site for the hybrid system units and SOPs devices simulated in NEPLAN. Optimal operation results in reducing system losses by 57% and by default reduced the branches flow congestion; also enhanced system voltage profile. As it is a new optimizer proposed for that problem solution; it goes into a statistical evaluation and convergence test. As a summary, this research work proved that technical, economic and environmental perspectives can be achieved together for a real case study, through decent planning of HRERs and optimal system operation utilizing SOP devices and distributed renewable sources of energy.

APPENDIX

See Table 7.

ACKNOWLEDGMENT

The authors would like to thank Professor/Araz who is a native speaker for his valuable review of our manuscript and also for enhancing English language structure of our work and the authors would also like to thank Professor/R. A. El-Sehiemy for his valuable review.

REFERENCES

- [1] V. Indragandhi, V. Subramaniaswamy, and R. Logesh, "Resources, configurations, and soft computing techniques for power management and control of PV/wind hybrid system," *Renew. Sustain. Energy Rev.*, vol. 69, pp. 129–143, Mar. 2017.
- [2] International Renewable Energy Agency. *Global Energy, a Roadmap to 2050 by Irena 2019 Edition*. Accessed: Nov. 10, 2019. [Online]. Available: <http://www.irena.org>
- [3] M. B. Shafik, G. I. Rashed, H. Chen, M. R. Elkadeem, and S. Wang, "Techno-economic design and optimal operation of active radial active distribution network using online optimizer: Real case study in Sohag, Egypt," in *Advances in Natural Computation, Fuzzy Systems and Knowledge Discovery (Advances in Intelligent Systems and Computing)*, vol. 1074. Cham, Switzerland: Springer, Jul. 2011, pp. 460–471, doi: [10.1007/978-3-030-32456-8_50](https://doi.org/10.1007/978-3-030-32456-8_50).
- [4] M. Elkadeem, S. Wang, E. G. Atia, M. Shafik, S. W. Sharshir, Z. Ullah, and H. Chen, "Techno-economic design and assessment of grid-isolated hybrid renewable energy system for agriculture sector," in *Proc. 14th IEEE Conf. Ind. Electron. Appl. (ICIEA)*, Jun. 2019, pp. 1562–1568, doi: [10.1109/iciea.2019.8834172](https://doi.org/10.1109/iciea.2019.8834172).
- [5] International Renewable Energy Agency. *Topics/Renewables/Renewables 2019 Report*. Accessed: Nov. 15, 2019. [Online]. Available: <https://www.irena.org>
- [6] A. Al-Sharafi, A. Z. Sahin, T. Ayar, and B. S. Yilbas, "Techno-economic analysis and optimization of solar and wind energy systems for power generation and hydrogen production in Saudi Arabia," *Renew. Sustain. Energy Rev.*, vol. 69, pp. 33–49, Mar. 2017.
- [7] International Renewable Energy Agency. *Renewable Energy in the Arab Region: Overview of Developments*. Accessed: Nov. 10, 2019. [Online]. Available: <https://www.irena.org/>
- [8] F. K. Abo-Elyousr and A. Elnozayh, "Bi-objective economic feasibility of hybrid micro-grid systems with multiple fuel options for islanded areas in Egypt," *Renew. Energy*, vol. 128, pp. 37–56, Dec. 2018.
- [9] S. Bahramara, M. P. Moghaddam, and M. Haghifam, "Optimal planning of hybrid renewable energy systems using HOMER: A review," *Renew. Sustain. Energy Rev.*, vol. 62, pp. 609–620, Sep. 2016.
- [10] R. Siddaiah and R. Saini, "A review on planning, configurations, modeling and optimization techniques of hybrid renewable energy systems for off grid applications," *Renew. Sustain. Energy Rev.*, vol. 58, pp. 376–396, May 2016.
- [11] A. Ehsan and Q. Yang, "State-of-the-art techniques for modelling of uncertainties in active distribution network planning: A review," *Appl. Energy*, vol. 239, pp. 1509–1523, Apr. 2019.
- [12] M. Usman, M. Coppo, F. Bignucolo, and R. Turri, "Losses management strategies in active distribution networks: A review," *Electr. Power Syst. Res.*, vol. 163, pp. 116–132, Oct. 2018.
- [13] F. Meng, B. Chowdhury, and M. S. Hossain, "Optimal integration of DER and SST in active distribution networks," *Int. J. Electr. Power Energy Syst.*, vol. 104, pp. 626–634, Jan. 2019.
- [14] F. Conte, F. D'Agostino, and F. Silvestro, "Operational constrained non-linear modeling and identification of active distribution networks," *Electr. Power Syst. Res.*, vol. 168, pp. 92–104, Mar. 2019.
- [15] J. M. Bloemink and T. C. Green, "Increasing distributed generation penetration using soft normally-open points," in *Proc. IEEE PES Gen. Meeting*, Jul. 2010, pp. 1–8, doi: [10.1109/pes.2010.5589629](https://doi.org/10.1109/pes.2010.5589629).
- [16] J. M. Bloemink and T. C. Green, "Increasing photovoltaic penetration with local energy storage and soft normally-open points," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Jul. 2011, pp. 1–8, doi: [10.1109/pes.2011.6039561](https://doi.org/10.1109/pes.2011.6039561).
- [17] L. Zhang, C. Shen, Y. Chen, S. Huang, and W. Tang, "Coordinated allocation of distributed generation, capacitor banks and soft open points in active distribution networks considering dispatching results," *Appl. Energy*, vol. 231, pp. 1122–1131, Dec. 2018.
- [18] J. M. Framinan and P. Perez-Gonzalez, "The 2-stage assembly flowshop scheduling problem with total completion time: Efficient constructive heuristic and metaheuristic," *Comput. Oper. Res.*, vol. 88, pp. 237–246, Dec. 2017.
- [19] M. B. Shafik, H. Chen, G. I. Rashed, and R. A. El-Sehiemy, "Adaptive multi objective parallel seeker optimization algorithm for incorporating TCSC devices into optimal power flow framework," *IEEE Access*, vol. 7, pp. 36934–36947, 2019, doi: [10.1109/access.2019.2905266](https://doi.org/10.1109/access.2019.2905266).
- [20] Anita and A. Yadav, "AEFA: Artificial electric field algorithm for global optimization," *Swarm Evol. Comput.*, vol. 48, pp. 93–108, Aug. 2019, doi: [10.1016/j.swevo.2019.03.013](https://doi.org/10.1016/j.swevo.2019.03.013).
- [21] T. Yuvaraj and K. Ravi, "Multi-objective simultaneous DG and DSTATCOM allocation in radial distribution networks using cuckoo searching algorithm," *Alexandria Eng. J.*, vol. 57, no. 4, pp. 2729–2742, Dec. 2018.
- [22] IEC International Standards. *Pre-Releases: IEC 60502-2:2014*. Accessed: Nov. 10, 2019. [Online]. Available: <https://webstore.iec.ch/>
- [23] W. Cao, J. Wu, N. Jenkins, C. Wang, and T. Green, "Operating principle of soft open points for electrical distribution network operation," *Appl. Energy*, vol. 164, pp. 245–257, Feb. 2016.
- [24] M. B. Shafik, H. Chen, G. I. Rashed, R. A. El-Sehiemy, M. R. Elkadeem, and S. Wang, "Adequate topology for efficient energy resources utilization of active distribution networks equipped with soft open points," *IEEE Access*, vol. 7, pp. 99003–99016, 2019, doi: [10.1109/access.2019.2930631](https://doi.org/10.1109/access.2019.2930631).
- [25] M. Shafik, G. Rashed, H. Chen, M. Elkadeem, and S. Wang, "Reconfiguration strategy for active distribution networks with soft open points," in *Proc. 14th IEEE Conf. Ind. Electron. Appl. (ICIEA)*, Xi'an, China, Jun. 2019.